

DSP — HW02 Cheat Sheet

Convolution sum: proofs, properties, index arithmetic & matrix form

Definition

DEF

Convolution sum

$$y[n] = (x * h)[n] = \text{sum over } k \text{ from } -\infty \text{ to } +\infty \text{ of } x[k] * h[n-k]$$

Interpretation: h is flipped and slid across x . Each output sample $y[n]$ is the dot product of x with the flipped, shifted version of h at position n .

Q1 — Time-invariance of the convolution sum

Q1

Prove convolution is time-invariant

Use the two-path test:

Path 1 — delay input first, then convolve:

$$T\{x[n-n_0]\} = \text{sum}_k x[k] * h[n-n_0-k]$$

Path 2 — convolve first, then delay output:

$$y[n-n_0] = \text{sum}_k x[k] * h[(n-n_0)-k] = \text{sum}_k x[k] * h[n-n_0-k]$$

Both paths give the same sum. Therefore convolution is time-invariant. [checkmark]

Q2 — Convolution properties (proofs)

Identity

$$x[n] * \delta[n] = x[n]$$

Key trick: $\delta[n-k] = 1$ only when $k = n$, zero otherwise.

$$\text{sum}_k x[k] * \delta[n-k] = x[n] * \delta[0] = x[n] * 1 = x[n] \text{ [checkmark]}$$

Delay

$$x[n] * \delta[n-n_0] = x[n-n_0]$$

Same trick: $\delta[n-n_0-k] = 1$ only when $k = n-n_0$.

The single surviving term gives $x[n-n_0]$. [checkmark]

Commutative

$$x[n] * h[n] = h[n] * x[n]$$

Start from: $\text{sum}_k x[k] * h[n-k]$

Substitute $m = n-k$ (so $k = n-m$, $dk \rightarrow -dm$, limits flip back to $-\infty..+\infty$):

$$= \text{sum}_m x[n-m] * h[m] = h[n] * x[n] \text{ [checkmark]}$$

Bilinear

$$x[n] * (a * h_1[n] + b * h_2[n]) = a * (x * h_1)[n] + b * (x * h_2)[n]$$

Expand inside the sum and use linearity of summation:

$$\text{sum}_k x[k] * (a * h_1[n-k] + b * h_2[n-k])$$

$$= a * \text{sum}_k x[k] * h_1[n-k] + b * \text{sum}_k x[k] * h_2[n-k]$$

$$= a * (x * h_1)[n] + b * (x * h_2)[n] \text{ [checkmark]}$$

Constants factor out of sums; sums split over addition.

Associative

$$(x * h_1) * h_2 = x * (h_1 * h_2)$$

Write out the double sum ($x * h_1$ substituted into the outer convolution):

$$\text{sum}_m [\text{sum}_k x[k] * h_1[m-k]] * h_2[n-m]$$

Substitute $l = m-k$ (so $m = l+k$):

$$= \text{sum}_k \text{sum}_l x[k] * h_1[l] * h_2[n-l-k]$$

Regroup: pull sum over l inside:

$$= \text{sum}_k x[k] * [\text{sum}_l h_1[l] * h_2[n-k-l]]$$

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This is  $(h_1 * h_2)[n-k]$

$$= \text{sum}_k x[k] * (h_1 * h_2)[n-k] = x * (h_1 * h_2) \text{ [checkmark]}$$

Key trick: substitute  $l = m-k$  to decouple the two dummy variables.

## Q3 — Finite sequence length & index arithmetic

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| <b>Q3a</b><br><b>Maximum length of <math>z = x * y</math> if <math>x</math> has length <math>k</math> and <math>y</math> has length <math>l</math></b>                             | <p>Answer: <math>k + l - 1</math></p> <p>Why: convolution starts when the first sample of <math>y</math> meets the first sample of <math>x</math>, and ends when the last sample of <math>y</math> passes the last sample of <math>x</math>. That gives <math>k + l - 1</math> output samples total.</p>                                                                                                                                                                                                                                                  |
| <b>Q3b</b><br><b>Index of <math>n=0</math> in <math>z</math>, given <math>n=0</math> is at array index <math>zx</math> in <math>x</math> and <math>zy</math> in <math>y</math></b> | <p>Answer: <math>zx + zy</math></p> <p>Why: the first output sample combines the first samples of both arrays. Their <math>n=0</math> positions simply add together in the output array.</p>                                                                                                                                                                                                                                                                                                                                                              |
| <b>Q3c</b><br><b><math>x[n] \neq 0</math> for <math>n=i..l</math>, <math>y[n] \neq 0</math> for <math>n=j..J</math>. First and last non-zero indices of <math>z</math>?</b>        | <p>First non-zero index of <math>z</math>: <math>i + j</math></p> <p>Last non-zero index of <math>z</math>: <math>l + J</math></p> <p>Why: the output is non-zero only when both <math>x</math> and <math>y</math> overlap.</p> <p>Earliest overlap: first sample of <math>x</math> (<math>n=i</math>) with first sample of <math>y</math> (<math>n=j</math>) <math>\rightarrow i+j</math></p> <p>Latest overlap: last sample of <math>x</math> (<math>n=l</math>) with last sample of <math>y</math> (<math>n=J</math>) <math>\rightarrow l+J</math></p> |

## Q4 — Matrix form of convolution

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| <b>Q4</b><br><b>Write <math>y = x * h</math> as <math>y = H * x</math> matrix multiplication</b> | <p><math>x = \{1, 2, 3, 2, 1\}</math> (length 5), <math>h = \{-1, 2, 1\}</math> (length 3), <math>y</math> has length <math>5+3-1 = 7</math></p> <p><math>H</math> is <math>7 \times 5</math>. Each row is <math>h</math> shifted right by one position.</p> <p>Start with <math>h</math> right-aligned in row 0, shift down until <math>h</math> is left-aligned in row 6:</p> <p><math>H = \begin{bmatrix} 1 &amp; 0 &amp; 0 &amp; 0 &amp; 0 \end{bmatrix}</math> &lt;- only last element of <math>h</math></p> <p><math>\begin{bmatrix} 2 &amp; 1 &amp; 0 &amp; 0 &amp; 0 \end{bmatrix}</math></p> <p><math>\begin{bmatrix} -1 &amp; 2 &amp; 1 &amp; 0 &amp; 0 \end{bmatrix}</math> &lt;- full <math>h</math></p> <p><math>\begin{bmatrix} 0 &amp; -1 &amp; 2 &amp; 1 &amp; 0 \end{bmatrix}</math></p> <p><math>\begin{bmatrix} 0 &amp; 0 &amp; -1 &amp; 2 &amp; 1 \end{bmatrix}</math></p> <p><math>\begin{bmatrix} 0 &amp; 0 &amp; 0 &amp; -1 &amp; 2 \end{bmatrix}</math></p> <p><math>\begin{bmatrix} 0 &amp; 0 &amp; 0 &amp; 0 &amp; -1 \end{bmatrix}</math> &lt;- only first element of <math>h</math></p> <p><math>y = H * x</math> gives the 7-element convolution output.</p> <p>Key insight: each row is a shifted version of <math>h</math> padded with zeros.</p> <p>This is why convolution is a LINEAR operation (matrix multiplication is linear).</p> |
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